

MECH 480: Finite Element Analysis and Solid Mechanics

Term: Fall 2026 | Instructor: Dr. Stephen Timoshenko (stimoshenko@engineering.univ.edu)

Course Description & Overview:

Theoretical formulations and computational implementations of 1D, 2D, and shell finite elements for linear elasticity and non-linear plasticity.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
1D Truss & Beam FEA Solver Implementation	25%	Week 04	MATLAB/Python Code & Verification
2D Plane Stress & Strain Ansys Analysis	30%	Week 08	Ansys FEA Simulation Report
Non-Linear Crashworthiness Simulation Capstone	35%	Week 12	Dynamic Crash Modeling Report
Engineering Ethics & Safety Brief	10%	Week 06	Safety Code of Ethics Brief

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Direct Stiffness Method & Truss	Logan (Ch. 1–3)
Week 02	Module 02: Strong & Weak Forms of Elasticity	Belytschko et al. (Ch. 1 & 2)
Week 03	Module 03: Variational Principles & Ritz Method	Reddy (Ch. 2–4)
Week 04	Module 04: Isoparametric Formulations & Quadrature	Hughes (Ch. 3)
Week 05	Module 05: 2D Plane Stress & Axisymmetric Elements	Cook et al. (Ch. 5–7)
Week 06	Module 06: Ansys APDL Scripting & Convergence	Ansys Mechanical Documentation
Week 07	Module 07: Plate and Shell Finite Elements	Timoshenko & Woinowsky-Krieger (Ch. 4)
Week 08	Module 08: Dynamic Structural Analysis & Modes	Bathe (Ch. 10 & 11)
Week 09	Module 09: Geometric Non-Linearity & Deformations	Belytschko (Ch. 6)
Week 10	Module 10: Material Non-Linearity & Plasticity	Simo & Hughes (Ch. 3)
Week 11	Module 11: Contact Mechanics & Formulations	Wriggers (Ch. 4)
Week 12	Module 12: Final Simulation Capstone Demos	Computational Mechanics Review